

Supplementary Material for “Confidence Intervals for Parameters in High-dimensional Sparse Vector Autoregression”

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The document provides detailed proofs of the theoretical results in the main text and additional simulation results.

1. Proofs of the theoretical results

Our proofs require several theoretical results from Zheng and Raskutti (2019).

5 Firstly, we bound the extreme eigenvalue of Σ , which can be obtained directly from Lemma 5.5 in Zheng and Raskutti (2019).

Proposition 1. *Under Assumption 1, we have*

$$1/\Lambda_{\min}(\Sigma) = O(1), \quad \Lambda_{\max}(\Sigma) = O(1).$$

Secondly, the following concentration inequalities were obtained from Lemma 5.2 in Zheng and Raskutti (2019).

Proposition 2. *Under Assumption 1,*

(a) *There exists a constant $c_1 > 0$ such that for any vector $v \in \mathbb{R}^K$ with $\|v\|_2 \leq 1$, for any $\eta \geq 0$, we have*

$$P[v^\top(\mathbf{X}^\top \mathbf{X}/n - \Sigma)v > Q_1 \eta] \leq c' \exp[-c_1 n \min\{\eta^2, \eta\}], \quad (1)$$

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where Q_1 is a constant depending only on the dependence structure of the VAR model.

(b) For $(K-1) \times K$ matrix W and K -dimensional vector w , if $\text{Cov}(W\mathbf{y}_t, w^T\mathbf{y}_t) = 0$ for every $t \in \mathbb{Z}$, then there exist constants c_2 such that for any vector $u \in \mathbb{R}^{K-1}$ with $\|u\|_2 \leq 1$, for any $\eta \geq 0$, we have

$$P[u^T(W\mathbf{X}^T\mathbf{X}w/n) > Q_2\eta] \leq c' \exp[-c_2n \min\{\eta^2, \eta\}], \quad (2)$$

where Q_2 is a constant depending only on the dependence structure of the VAR model.

Thirdly, the following Proposition 3 were obtained from Lemma A.1. and Lemma 5.1 in Zheng and Raskutti (2019). The first inequality is related to the restricted eigenvalue (RE) condition for the design matrix $\mathbf{X}$ and we modify the original conclusion to fit our proof related to the nodewise Lasso. The second

15 is the deviation bound for $\mathbf{X}^T\varepsilon_i/n$.

Proposition 3. (a) Under Assumptions 1, 3 and 4, we have, in probability,

$$\theta'(\mathbf{X}^T\mathbf{X}/n)\theta \geq \alpha\|\theta\|_2^2 - \tau\|\theta\|_1^2, \quad \forall \theta \in \mathbb{R}^K, \quad (3)$$

where $\alpha > 0$ is a constant depending only on the dependence structure of the VAR model, and τ satisfies $\tau s_i = O(1)$ for any i and $\tau q_j = O(1)$ for any j .

(b) Under Assumptions 1 and 3, for $i = 1, \dots, K$, we have, in probability,

$$\|\mathbf{X}^T\varepsilon_i/n\|_\infty \leq C_1\sqrt{\log(K)/n}, \quad (4)$$

where $C_1 > 0$ is a constant depending only on the dependence structure of the VAR model.

Lastly, to prove the sparse Riesz condition, we need Lemma F.2 in the supplementary material of Basu and Michailidis (2015), which used discretization

20 to expand the bound of a single vector in (1) to the bound of a set of sparse vectors.

Lemma 1. Consider a symmetric matrix $D_{K \times K}$. If, for any vector $v \in \mathbb{R}^K$ with $\|v\|_2 \leq 1$, and any $\eta \geq 0$,

$$P[v^\top Dv > C\eta] \leq c' \exp[-cn \min\{\eta^2, \eta\}],$$

then, for any integer $s \geq 1$, we have

$$\begin{aligned} P\left[\sup_{\|v\|_0 \leq s} \sup_{\|v\|_2 \leq 1} |v^\top Dv| > C\eta\right] \\ \leq c' \exp[-cn \min\{\eta, \eta^2\} + s \min\{\log(K), \log(21eK/s)\}]. \end{aligned}$$

1.1. Proofs of the theoretical results in Section 3.1

We first show that the following sparse Riesz condition (Zhang and Huang, 2008) holds with probability tending to 1:

$$c_* \leq \min_{\|v\|_0 \leq s_R} \min_{\|v\|_2=1} \|\mathbf{X}v\|_2^2/n \leq \max_{\|v\|_0 \leq s_R} \max_{\|v\|_2=1} \|\mathbf{X}v\|_2^2/n \leq c^*. \quad (5)$$

Proof of Proposition 1. By (1) in Proposition 2 and Lemma 1, we have

$$\begin{aligned} P\left[\sup_{\|v\|_0 \leq s_R} \sup_{\|v\|_2 \leq 1} |v^\top (\mathbf{X}^\top \mathbf{X}/n - \Sigma)v| > Q_1 \eta\right] \leq c' \exp[-cn \min\{\eta, \eta^2\} \\ + s_R \min\{\log(K), \log(21eK/s_R)\}]. \end{aligned}$$

Thus, we have

$$\begin{aligned} & \sup_{\|v\|_0 \leq s_R} \sup_{\|v\|_2 \leq 1} |v^\top (\mathbf{X}^\top \mathbf{X}/n - \Sigma)v| \\ &= O_p(\sqrt{s_R \log(K)/n}) \\ &= O_p(\sqrt{s_i \log(K)/n}) \\ &= o_p(1), \end{aligned}$$

where the last equality is due to the sparsity Assumption 3 ($s_i \log(K)/\sqrt{n} \rightarrow 0$).

According to Proposition 1, we have

$$\sup_{\|v\|_0 \leq s_R} \sup_{\|v\|_2 \leq 1} |v^\top \Sigma v| = O(1), \quad 1/\sup_{\|v\|_0 \leq s_R} \sup_{\|v\|_2 \leq 1} |v^\top \Sigma v| = O(1).$$

The result follows from the triangle inequality. $\square$

Now, we can prove Theorem 1. As statement (a) has been obtained by Zheng and Raskutti (2019) (see Lemma 3.1), we need only to prove the statement (b).

Proof of Theorem 1 (b). Firstly, we introduce some notations. We define the following sets:

$$\begin{aligned}
A_{tp} &:= \{j : a_{ij} \neq 0\} \cap \{j : \hat{a}_{ij}^{\text{Lasso}} \neq 0\}, \\
A_{fp} &:= \{j : a_{ij} = 0\} \cap \{j : \hat{a}_{ij}^{\text{Lasso}} \neq 0\}, \\
A_{fn} &:= \{j : a_{ij} \neq 0\} \cap \{j : \hat{a}_{ij}^{\text{Lasso}} = 0\}, \\
A_{tn} &:= \{j : a_{ij} = 0\} \cap \{j : \hat{a}_{ij}^{\text{Lasso}} = 0\}, \\
A_1 &:= A_{tp} \cup A_{fp} \cup A_{fn}, \quad A_2 := A_{tp} \cup A_{fn}, \quad A_3 := A_{fp},
\end{aligned}$$

where the subscripts represent true positive, false positive, false negative, and true negative, respectively. Note that we omit the subscript i of these sets for simplicity. Let

$$\hat{s}_+ := |A_1|.$$

According to these definitions, we have

$$s_i = |A_2| \leq |A_1| = \hat{s}_+, \quad \hat{s}_i = |A_{tp} \cup A_{fp}| \leq |A_1| = \hat{s}_+, \quad |A_3| = \hat{s}_+ - s_i. \quad (6)$$

Clearly, $A_2 \subseteq A_1$ and $A_3 \subseteq A_1$. For $k = 2, 3$, let $\mathbf{Q}_{k1}$ be an $|A_k| \times |A_1|$ transformation matrix with elements 0 and 1. $\mathbf{Q}_{k1}$ selects variables in A_k from A_1 , such that $\mathbf{Q}_{k1} \mathbf{X}_{A_1}^T = \mathbf{X}_{A_k}^T$. Recall that

$$\hat{a}_i^{\text{Lasso}} := \underset{\alpha \in \mathbb{R}^K}{\operatorname{argmin}} \{ \|Y_i - \mathbf{X}\alpha\|_2^2/n + 2\lambda \|\alpha\|_1 \},$$

where $\hat{a}_i^{\text{Lasso}} = (\hat{a}_{i1}^{\text{Lasso}}, \dots, \hat{a}_{iK}^{\text{Lasso}})$. By KKT condition, we have

$$\begin{cases} X_j^T(Y_i - \mathbf{X}\hat{a}_i^{\text{Lasso}})/n = \lambda \operatorname{sign}(\hat{a}_{ij}^{\text{Lasso}}), & \hat{a}_{ij}^{\text{Lasso}} \neq 0 \\ |X_j^T(Y_i - \mathbf{X}\hat{a}_i^{\text{Lasso}})/n| \leq \lambda, & \hat{a}_{ij}^{\text{Lasso}} = 0 \end{cases}. \quad (7)$$

For $k = 2, 3$, we denote

$$\xi_k := \mathbf{X}_{A_k}^T(Y_i - \mathbf{X}\hat{a}_i^{\text{Lasso}})/(n\lambda), \quad (8)$$

and

$$\mathbf{v}_k := \lambda\sqrt{n} \hat{\Sigma}_{A_1, A_1}^{-1/2} \mathbf{Q}_{k1}^T \xi_k, \quad (9)$$

where $\hat{\Sigma}_{A_1, A_1} := \mathbf{X}_{A_1}^T \mathbf{X}_{A_1} / n$. By (7), we have

$$\|\xi_2\|_2^2 \leq |A_2|, \quad \|\xi_3\|_2^2 = |A_3| = \hat{s}_+ - s_i. \quad (10)$$

Next, we intend to proving that $\hat{s}_+ = O_p(s_i)$, which implies $\hat{s}_i = O_p(s_i)$ by (6). Our proof is divided into three steps. In steps 1 and 2, taking $\mathbf{v}_3$ as a bridge, we prove that $\hat{s}_+ \leq s_R$ implies $\hat{s}_+ \leq (2 + 2c^*/c_*)s_i$. Specifically, we give a lower bound and an upper bound of $\|\mathbf{v}_3\|_2^2$ in steps 1 and 2 respectively. In step 3, using the results in steps 1 and 2, we prove the desired result by contradiction.

Step 1. Assuming that $\{\hat{s}_+ \leq s_R\}$, by (5) and (10), we have

$$\begin{aligned} \|\mathbf{v}_3\|_2^2 &= \lambda^2 n \xi_3^T \mathbf{Q}_{31} \hat{\Sigma}_{A_1, A_1}^{-1} \mathbf{Q}_{31}^T \xi_3 \\ &\geq \lambda^2 n \|\mathbf{Q}_{31}^T \xi_3\|_2^2 / c^* \\ &= \lambda^2 n (\hat{s}_+ - s_i) / c^*. \end{aligned} \quad (11)$$

Step 2. Assuming $\{\hat{s}_+ \leq s_R\}$, as $a_{i, A_{tn}} = \hat{a}_{i, A_{tn}}^{\text{Lasso}} = \mathbf{0}$ and $A_2 \cup A_3 = A_1$, we have

$$\begin{aligned} n\lambda(\mathbf{Q}_{21}^T \xi_2 + \mathbf{Q}_{31}^T \xi_3) &= \mathbf{X}_{A_1}^T (Y_i - \mathbf{X}_{A_1} \hat{a}_i^{\text{Lasso}}) \\ &= \mathbf{X}_{A_1}^T (Y_i - \mathbf{X}_{A_1} \hat{a}_{i, A_1}^{\text{Lasso}}) \\ &= \mathbf{X}_{A_1}^T \mathbf{X}_{A_1} a_{i, A_1} + \mathbf{X}_{A_1}^T \mathbf{X}_{A_{tn}} a_{i, A_{tn}} + \mathbf{X}_{A_1}^T \varepsilon_i - \mathbf{X}_{A_1}^T \mathbf{X}_{A_1} \hat{a}_{i, A_1}^{\text{Lasso}} \\ &= n\hat{\Sigma}_{A_1, A_1} a_{i, A_1} + \mathbf{X}_{A_1}^T \varepsilon_i - n\hat{\Sigma}_{A_1, A_1} \hat{a}_{i, A_1}^{\text{Lasso}}. \end{aligned} \quad (12)$$

Then we have

$$\begin{aligned} \mathbf{v}_3^T (\mathbf{v}_2 + \mathbf{v}_3) &= \lambda^2 n \xi_3^T \mathbf{Q}_{31} \hat{\Sigma}_{A_1, A_1}^{-1} (\mathbf{Q}_{21}^T \xi_2 + \mathbf{Q}_{31}^T \xi_3) \\ &= \lambda \xi_3^T \mathbf{Q}_{31} \hat{\Sigma}_{A_1, A_1}^{-1} \mathbf{X}_{A_1}^T \varepsilon_i + \lambda n \xi_3^T \mathbf{Q}_{31} (a_{i, A_1} - \hat{a}_{i, A_1}^{\text{Lasso}}) \\ &\leq \lambda \xi_3^T \mathbf{Q}_{31} \hat{\Sigma}_{A_1, A_1}^{-1} \mathbf{X}_{A_1}^T \varepsilon_i \\ &\leq \lambda n \|\xi_3^T \mathbf{Q}_{31} \hat{\Sigma}_{A_1, A_1}^{-1}\|_1 \|\mathbf{X}_{A_1}^T \varepsilon_i / n\|_\infty, \end{aligned} \quad (13)$$

where the first inequality is due to

$$\mathbf{Q}_{31} (a_{i, A_1} - \hat{a}_{i, A_1}^{\text{Lasso}}) = a_{i, A_3} - \hat{a}_{i, A_3}^{\text{Lasso}} = -\hat{a}_{i, A_3}^{\text{Lasso}},$$

and

$$\xi_3^T \hat{a}_{i,A_3}^{\text{Lasso}} \geq 0,$$

by the KKT condition (7), and the second inequality is due to Hölder inequality. Furthermore, we have

$$\begin{aligned} \|\xi_3^T \mathbf{Q}_{31} \hat{\Sigma}_{A_1, A_1}^{-1}\|_1 &\leq \sqrt{\hat{s}_+} \|\xi_3^T \mathbf{Q}_{31} \hat{\Sigma}_{A_1, A_1}^{-1}\|_2 \\ &\leq \sqrt{\hat{s}_+} \|\xi_3^T \mathbf{Q}_{31}\|_2 / c_* \\ &\leq \hat{s}_+ / c_*, \end{aligned} \tag{14}$$

where the second inequality is due to the sparse Riesz condition (5), and the last inequality is because $\|\xi_3^T \mathbf{Q}_{31}\|_2 = \|\xi_3\|_2 = \sqrt{\hat{s}_+ - s_i}$. By sparse Riesz condition (5) and (10), we also have

$$\begin{aligned} \|\mathbf{v}_2\|_2^2 &= \lambda^2 n \xi_2^T \mathbf{Q}_{21} \hat{\Sigma}_{A_1, A_1}^{-1} \mathbf{Q}_{21}^T \xi_2 \\ &\leq \lambda^2 n \|\mathbf{Q}_{21}^T \xi_2\|_2^2 / c_* \\ &\leq \lambda^2 n s_i / c_*. \end{aligned} \tag{15}$$

By (13), triangle inequality, Hölder inequality, (14) and (15), we have

$$\begin{aligned} \|\mathbf{v}_3\|_2^2 &\leq \lambda n \|\xi_3^T \mathbf{Q}_{31} \hat{\Sigma}_{A_1, A_1}^{-1}\|_1 \|\mathbf{X}_{A_1}^T \varepsilon_i / n\|_\infty + \|\mathbf{v}_2\|_2 \|\mathbf{v}_3\|_2 \\ &\leq (\lambda n \hat{s}_+ / c_*) \|\mathbf{X}_{A_1}^T \varepsilon_i / n\|_\infty + (\lambda^2 n s_i / c_*)^{1/2} \|\mathbf{v}_3\|_2. \end{aligned} \tag{16}$$

Now we need to bound the term $\|\mathbf{X}_{A_1}^T \varepsilon_i / n\|_\infty$. By (4), we have, in probability,

$$\|\mathbf{X}_{A_1}^T \varepsilon_i / n\|_\infty \leq \|\mathbf{X}^T \varepsilon_i / n\|_\infty \leq C_1 \sqrt{\log(K)/n}. \tag{17}$$

As $x^2 \leq c + 2bx$ implies $x^2 \leq (b + \sqrt{b^2 + c})^2 \leq 2c + 4b^2$ for $x = \|\mathbf{v}_3\|_2$, by (16) and (17), we have, in probability,

$$\|\mathbf{v}_3\|_2^2 \leq 2(\lambda n \hat{s}_+ / c_*) C_1 \sqrt{\log(K)/n} + \lambda^2 n s_i / c_*. \tag{18}$$

Combining with the lower bound (11), we have

$$\lambda^2 n (\hat{s}_+ - s_i) / c_* \leq 2(\lambda n \hat{s}_+ / c_*) C_1 \sqrt{\log(K)/n} + \lambda^2 n s_i / c_*.$$

Moving the terms related to $\hat{s}_+$ and s_i to each side of the inequality, we have

$$\left(\lambda - \frac{2C_1 c^*}{c_*} \sqrt{\frac{\log(K)}{n}} \right) \hat{s}_+ \leq \left(1 + \frac{c^*}{c_*} \right) \lambda s_i.$$

If

$$\lambda \geq \frac{4C_1 c^*}{c_*} \sqrt{\frac{\log(K)}{n}},$$

we have

$$\frac{1}{2} \lambda \hat{s}_+ \leq \left(\lambda - \frac{2C_1 c^*}{c_*} \sqrt{\frac{\log(K)}{n}} \right) \hat{s}_+ \leq \left(1 + \frac{c^*}{c_*} \right) \lambda s_i,$$

then

$$\hat{s}_+ \leq (2 + 2c^*/c_*) s_i.$$

Thus, we obtain that

$$\hat{s}_+ \leq s_R \quad \Rightarrow \quad \hat{s}_+ \leq (2 + 2c^*/c_*) s_i. \quad (19)$$

Step 3. Note that $\hat{s}_+ = \hat{s}_+(\lambda)$ is a function of λ . We denote

$$\hat{s}_{+, \min} := \min_{\lambda \geq 4C_1 c^*/c_* \sqrt{\log(K)/n}} \hat{s}_+(\lambda), \quad \hat{s}_{+, \max} := \max_{\lambda \geq 4C_1 c^*/c_* \sqrt{\log(K)/n}} \hat{s}_+(\lambda).$$

Because of the continuity of the Lasso path, we could choose the variable one-at-a-time. Therefore, when $\lambda \geq 4C_1 c^*/c_* \sqrt{\log(K)/n}$, $\hat{s}_+(\lambda)$ could take every integer from $\hat{s}_{+, \min}$ to $\hat{s}_{+, \max}$.

When $\lambda = \infty$, we have $A_1 = A_2$ and thus

$$\hat{s}_{+, \min} = s_i \leq (2 + 2c^*/c_*) s_i.$$

If $\hat{s}_{+, \max} > s_R$, by the definition of s_R in Proposition 1, we have

$$\hat{s}_{+, \max} > s_R > (2 + 2c^*/c_*) s_i + 1.$$

35 Then, there exists a $\lambda \geq 4C_1 c^*/c_* \sqrt{\log(K)/n}$, such that $\hat{s}_+ = \lfloor (2 + 2c^*/c_*) s_i + 1 \rfloor$, which contradicts (19) since $\hat{s}_+ \leq s_R$ and $\hat{s}_+ > (2 + 2c^*/c_*) s_i$. By contradiction, $\hat{s}_{+, \max} \leq s_R$ and our result follows. $\square$

Next, we prove the estimation and prediction error bounds of the nodewise Lasso estimator defined in (7). These bounds are useful in the proof of Theorem

2. Recall that these bounds are

$$\|\hat{\gamma}_j - \gamma_j\|_1 = O_p \left(q_j \sqrt{\log(K)/n} \right), \quad (20)$$

$$\|\mathbf{X}_{-j} (\hat{\gamma}_j - \gamma_j)\|_2^2/n = O_p (q_j \log(K)/n), \quad (21)$$

where $j = 1, \dots, K$.

Proof of Proposition 2. For $j = 1, \dots, K$, as $\hat{\gamma}_j$ is the minimizer:

$$\hat{\gamma}_j := \operatorname{argmin}_{\gamma \in \mathbb{R}^{K-1}} \{ \|X_j - \mathbf{X}_{-j}\gamma\|_2^2/n + 2\lambda_j \|\gamma\|_1 \},$$

we obtain the basic inequality

$$\|X_j - \mathbf{X}_{-j}\hat{\gamma}_j\|_2^2/n + 2\lambda_j \|\hat{\gamma}_j\|_1 \leq \|X_j - \mathbf{X}_{-j}\gamma_j\|_2^2/n + 2\lambda_j \|\gamma_j\|_1.$$

We denote $\delta := \hat{\gamma}_j - \gamma_j$. By simple algebra, we have

$$\|\mathbf{X}_{-j}\delta\|_2^2/n \leq 2\tilde{\varepsilon}^T \mathbf{X}_{-j}\delta/n + 2\lambda_j (\|\gamma_j\|_1 - \|\hat{\gamma}_j\|_1), \quad (22)$$

where $\tilde{\varepsilon} := X_j - \mathbf{X}_{-j}\gamma_j$. Since $\operatorname{Cov}(\mathbf{X}_{-j}, \tilde{\varepsilon}) = 0$, then by (2), we have

$$P(\|\tilde{\varepsilon}^T \mathbf{X}_{-j}\|_\infty > Q_2 \eta) \leq c' K \exp[-c_3 n \min\{\eta, \eta^2\}].$$

With $\eta = c_0 \sqrt{\log(K)/n}$ and suitable choice of $\lambda_j \asymp \sqrt{\log(K)/n}$, we have, in probability,

$$\|\tilde{\varepsilon}^T \mathbf{X}_{-j}\|_\infty \leq Q_2 c_0 \sqrt{\log(K)/n} \leq \lambda_j/2.$$

Furthermore, by Hölder inequality, we have

$$\tilde{\varepsilon}^T \mathbf{X}_{-j}\delta/n \leq \|\tilde{\varepsilon}^T \mathbf{X}_{-j}\|_\infty \|\delta\|_1 \leq \lambda_j \|\delta\|_1/2.$$

We denote the support set of γ_j by $\mathcal{K}$. By triangle inequality and $\gamma_{j\mathcal{K}^c} = 0$, we have

$$\|\gamma_j\|_1 - \|\hat{\gamma}_j\|_1 = \|\gamma_{j\mathcal{K}}\|_1 - \|\hat{\gamma}_{j\mathcal{K}}\|_1 + \|\gamma_{j\mathcal{K}^c}\|_1 - \|\hat{\gamma}_{j\mathcal{K}^c}\|_1 \leq \|\delta_{\mathcal{K}}\|_1 - \|\delta_{\mathcal{K}^c}\|_1.$$

Therefore, (22) becomes

$$\begin{aligned} 0 &\leq \|\mathbf{X}_{-j}\delta\|_2^2/n \leq \lambda_j \|\delta\|_1 + 2\lambda_j (\|\delta_{\mathcal{K}}\|_1 - \|\delta_{\mathcal{K}^c}\|_1) \\ &= \lambda_j (\|\delta_{\mathcal{K}}\|_1 + \|\delta_{\mathcal{K}^c}\|_1) + 2\lambda_j (\|\delta_{\mathcal{K}}\|_1 - \|\delta_{\mathcal{K}^c}\|_1) \\ &= 3\lambda_j \|\delta_{\mathcal{K}}\|_1 - \lambda_j \|\delta_{\mathcal{K}^c}\|_1 \\ &\leq 3\lambda_j \|\delta\|_1. \end{aligned} \quad (23)$$

Inequality (23) implies $\|\delta_{\mathcal{K}^c}\|_1 \leq 3\|\delta_{\mathcal{K}}\|_1$ so that

$$\|\delta\|_1 \leq 4\|\delta_{\mathcal{K}}\|_1 \leq 4\sqrt{q_j} \|\delta\|_2. \quad (24)$$

By (3) and (24), we have

$$\|\mathbf{X}_{-j}\delta\|_2^2/n \geq \alpha\|\delta\|_2^2 - \tau\|\delta\|_1^2 \geq \left(\frac{\alpha}{4q_j} - \tau\right)\|\delta\|_1^2. \quad (25)$$

By (23), (25), $\tau q_j = O(1)$ and Assumption 2 ($\lambda_j \asymp \sqrt{\log(K)/n}$), we obtain
 40 (20). Then by (23), (20) and Assumption 2, we obtain (21). $\square$

To prove Theroem 2, we denote the nodewise Lasso residuals by

$$\hat{Z}_j = (\hat{Z}_{1j}, \dots, \hat{Z}_{nj}) := X_j - \mathbf{X}_{-j}\hat{\gamma}_j,$$

which is an estimator of

$$Z_j = (Z_{1j}, \dots, Z_{nj}) := X_j - \mathbf{X}_{-j}\gamma_j.$$

Moreover, we denote

$$\tau_j^2 := E(\|Z_j\|_2^2/n).$$

Unlike the proof for high-dimensional sparse linear regression models, we should distinguish $\hat{Z}_j$ and Z_j carefully because of the correlation between $\mathbf{X}_{-j}$ and $\hat{\gamma}_j$.

Proof of Theorem 2. Recall that

$$\hat{a}_i - a_i = \hat{\Theta}\mathbf{X}^T\varepsilon_i/n + (I - \hat{\Theta}\hat{\Sigma})(\hat{a}_i^{\text{Lasso}} - a_i). \quad (26)$$

We prove the theorem in three steps. Step 1 proves that the second term of the right-hand of (26) is asymptotically negligible. Step 2 proves that the
 45 first term is asymptotically normal. Step 3 proves that our variance estimator, $\hat{\sigma}_i^2 = \|\hat{\varepsilon}_i\|_2^2/(n - \hat{s}_{i0})$, is consistent.

Step 1. Recall that

$$\hat{C} := \begin{bmatrix} 1 & -\hat{\gamma}_{1,2} & \cdots & -\hat{\gamma}_{1,K} \\ -\hat{\gamma}_{2,1} & 1 & \cdots & -\hat{\gamma}_{2,K} \\ \vdots & \vdots & \ddots & \vdots \\ -\hat{\gamma}_{K,1} & -\hat{\gamma}_{K,2} & \cdots & 1 \end{bmatrix},$$

$$\hat{\tau}_j^2 := \|X_j - \mathbf{X}_{-j}\hat{\gamma}_j\|_2^2/n + \lambda_j \|\hat{\gamma}_j\|_1, \quad \hat{T}^2 := \text{diag}(\hat{\tau}_1^2, \dots, \hat{\tau}_K^2), \quad \hat{\Theta} := \hat{T}^{-2}\hat{C}.$$

We denote the j th row of $\hat{\Theta}$ by $\hat{\Theta}_j$ and obtain

$$\mathbf{X}\hat{\Theta}_j^T/n = (X_j - \mathbf{X}_{-j}\hat{\gamma}_j)/(n\hat{\tau}_j^2). \quad (27)$$

According to the KKT conditions for the nodewise Lasso, we have

$$-\mathbf{X}_{-j}^T(X_j - \mathbf{X}_{-j}\hat{\gamma}_j)/n + \lambda_j\hat{\kappa}^j = 0, \quad (28)$$

where $\hat{\kappa}^j$ is the sub-gradient the of ℓ_1 norm evaluated at $\hat{\gamma}_j$ and satisfies $\|\hat{\kappa}^j\|_\infty \leq 1$ and $\hat{\kappa}_k^j = \text{sign}(\hat{\gamma}_{jk})$ if $\hat{\gamma}_{jk} \neq 0$. Multiplying both hand sides of (28) by $\hat{\gamma}_j$, we obtain

$$\lambda_j\|\hat{\gamma}_j\| = \hat{\gamma}_j^T\mathbf{X}_{-j}^T(X_j - \mathbf{X}_{-j}\hat{\gamma}_j)/n.$$

Substituting the above formula into the definition of $\hat{\tau}_j^2$, we have

$$\hat{\tau}_j^2 := \|X_j - \mathbf{X}_{-j}\hat{\gamma}_j\|_2^2/n + \lambda_j\|\hat{\gamma}_j\|_1 = X_j^T(X_j - \mathbf{X}_{-j}\hat{\gamma}_j)/n.$$

Combining with (27), we have

$$X_j^T\mathbf{X}\hat{\Theta}_j^T/n = X_j^T(X_j - \mathbf{X}_{-j}\hat{\gamma}_j)/(n\hat{\tau}_j^2) = 1. \quad (29)$$

By (27) and (28), we have

$$\|\mathbf{X}_{-j}^T\mathbf{X}\hat{\Theta}_j^T/n\|_\infty = \|\mathbf{X}_{-j}^T(X_j - \mathbf{X}_{-j}\hat{\gamma}_j)/(n\hat{\tau}_j^2)\|_\infty \leq \lambda_j/\hat{\tau}_j^2. \quad (30)$$

By (29) and (30), we have

$$\|(I - \hat{\Theta}\hat{\Sigma})\|_\infty = \max_j \|(e_j - \mathbf{X}^T\mathbf{X}\hat{\Theta}_j^T/n)\|_\infty \leq \max_j \lambda_j/\hat{\tau}_j^2.$$

By Hölder inequality, we have

$$\begin{aligned} \sqrt{n}\|(I - \hat{\Theta}\hat{\Sigma})(\hat{a}_i^{\text{Lasso}} - a_i)\|_\infty &\leq \sqrt{n}\|(I - \hat{\Theta}\hat{\Sigma})\|_\infty\|\hat{a}_i^{\text{Lasso}} - a_i\|_1 \\ &\leq \sqrt{n}\left(\max_j \lambda_j/\hat{\tau}_j^2\right)\|\hat{a}_i^{\text{Lasso}} - a_i\|_1. \end{aligned} \quad (31)$$

By Theorem 1, we have

$$\|\hat{a}_i^{\text{Lasso}} - a_i\|_1 = O_p(s_i\sqrt{\log(K)/n}).$$

Then, we need only to prove that for $j = 1, \dots, K$,

$$1/\hat{\tau}_j^2 = O_p(1). \quad (32)$$

As the proofs are the same for different j , for simplicity, we only consider $j = 1$. By simple algebra, we have the following bound for τ_1^2 , which is the population level counterpart of $\hat{\tau}_1^2$,

$$\begin{aligned} 1/\tau_1^2 &= \Theta_{11} = e_1^\top \Sigma^{-1} e_1 \leq 1/\Lambda_{\min}(\Sigma), \\ \tau_1^2 &\leq E[X_{11}]^2 = \Sigma_{1,1} = e_1^\top \Sigma e_1 \leq \Lambda_{\max}(\Sigma), \end{aligned} \quad (33)$$

where $1/\Lambda_{\min}(\Sigma) = O(1)$ and $\Lambda_{\max}(\Sigma) = O(1)$ by Proposition 1. By the definition and Cauchy–Schwarz inequality, we have,

$$\begin{aligned} |\hat{\tau}_1^2 - \tau_1^2| &\leq \left| \|\hat{Z}_1\|_2^2/n - \tau_1^2 \right| + \lambda_1 \|\hat{\gamma}_1\|_1 \\ &\leq \left| \|Z_1\|_2^2/n - \tau_1^2 \right| + \|\hat{Z}_1 - Z_1\|_2^2/n \\ &\quad + 2\|Z_1\|_2 \|\hat{Z}_1 - Z_1\|_2/n + \lambda_1 \|\hat{\gamma}_1\|_1. \end{aligned} \quad (34)$$

As $\|\hat{Z}_1 - Z_1\|_2^2/n = \|\mathbf{X}_{-1}(\hat{\gamma}_1 - \gamma_1)\|_2^2/n = O_p(q_1 \log(K)/n)$ has been bounded by (21), we need only to bound $|\|Z_1\|_2^2/n - \tau_1^2|$ and $\|\hat{\gamma}_1\|_1$.

Before that, we first bound the corresponding population level terms $\|\gamma_1\|_2$ and $\|\gamma_1\|_1$. Recall that $\gamma_1 = \Sigma_{-1,-1}^{-1} \Sigma_{21}$. We partition the covariance matrix Σ as

$$\Sigma = \begin{bmatrix} \Sigma_{1,1} & \Sigma_{1,-1} \\ \Sigma_{-1,1} & \Sigma_{-1,-1} \end{bmatrix}.$$

We denote

$$\begin{aligned} \Sigma_{11.2} &:= \Sigma_{1,1} - \Sigma_{1,-1} \Sigma_{-1,-1}^{-1} \Sigma_{-1,1} \\ &= \Sigma_{1,1} - \gamma_1^\top \Sigma_{-1,-1} \gamma_1. \end{aligned}$$

As $|\Sigma_{11.2}| |\Sigma_{-1,-1}| = |\Sigma| > 0$, $|\Sigma_{-1,-1}| > 0$ and $\Sigma_{11.2}$ is an one by one matrix (thus its determinant is itself), we have $\Sigma_{1,1} - \gamma_1^\top \Sigma_{-1,-1} \gamma_1 > 0$. Therefore,

$$\Sigma_{1,1} \geq \gamma_1^\top \Sigma_{-1,-1} \gamma_1 \geq \|\gamma_1\|_2^2 \Lambda_{\min}(\Sigma_{-1,-1}) \geq \|\gamma_1\|_2^2 \Lambda_{\min}(\Sigma).$$

Then we obtain

$$\|\gamma_1\|_2 \leq \sqrt{\Sigma_{1,1}/\Lambda_{\min}(\Sigma)} = O(1). \quad (35)$$

By Cauchy–Schwarz inequality,

$$\|\gamma_1\|_1 \leq \sqrt{\|\gamma_1\|_0} \|\gamma_1\|_2 \leq \sqrt{q_1} \|\gamma_1\|_2 \leq \sqrt{q_1 \Sigma_{1,1}/\Lambda_{\min}(\Sigma)} = O(\sqrt{q_1}). \quad (36)$$

Next, we study the term $\|Z_1\|_2^2/n$. With

$$\tilde{v} := (1, -\hat{\gamma}_{1,2}, \dots, -\hat{\gamma}_{1,K})^\top / \sqrt{\|\gamma_1\|_2^2 + 1},$$

we have

$$\begin{aligned} |\|Z_1\|_2^2/n - \tau_1^2| &= |\|X_1 - \mathbf{X}_{-1}\gamma_1\|_2^2/n - E(\|X_1 - \mathbf{X}_{-1}\gamma_1\|_2^2/n)| \\ &= (\|\gamma_1\|_2^2 + 1) |\tilde{v}^\top (\mathbf{X}^\top \mathbf{X}/n - \Sigma) \tilde{v}| \\ &= O_p(\sqrt{\log(K)/n}) \\ &= o_p(1), \end{aligned} \tag{37}$$

where the first equality is because of the definitions of $\|Z_1\|_2^2$ and τ_1^2 , the second is because of the definition of $\tilde{v}$, the third is because of (35) and (1) with $\eta = c_0 \sqrt{\log(K)/n}$, and the fourth is because of Assumption 3.

Now we can bound $\|\hat{\gamma}_1\|_1$. We have

$$\begin{aligned} \lambda_1 \|\hat{\gamma}_1\|_1 &\leq \lambda_1 \|\gamma_1\|_1 + \lambda_1 \|\hat{\gamma}_1 - \gamma_1\|_1 \\ &= O_p\left(\sqrt{q_1 \log(K)/n}\right) + O_p(q_1 \log(K)/n), \end{aligned} \tag{38}$$

where the first equality is due to triangle inequality and the second is due to (36), (20) and Assumption 2.

We continue to bound (34) with the results (21), (37), (33), (38) and Assumption 4,

$$\begin{aligned} |\hat{\tau}_1^2 - \tau_1^2| &\leq |\|Z_1\|_2^2/n - \tau_1^2| + \|\hat{Z}_1 - Z_1\|_2^2/n \\ &\quad + 2\|Z_1\|_2 \|\hat{Z}_1 - Z_1\|_2/n + \lambda_1 \|\hat{\gamma}_1\|_1 \\ &= O_p(q_1 \log(K)/n) + O_p(1) O_p\left(\sqrt{q_1 \log(K)/n}\right) \\ &\quad + O_p\left(\sqrt{q_1 \log(K)/n}\right) + O_p(q_1 \log(K)/n) \\ &= o_p(1). \end{aligned} \tag{39}$$

Combining with (33), we obtain (32). We summarize some intermediate results we obtain above as follows,

$$|\|Z_j\|_2^2/n - \tau_j^2| = o_p(1), \tag{40}$$

$$|\hat{\tau}_j^2 - \tau_j^2| = o_p(1). \tag{41}$$

By (31), (32) and Assumptions 2 and 3, we have

$$\sqrt{n}\|(I - \hat{\Theta}\hat{\Sigma})(\hat{a}_i^{\text{Lasso}} - a_i)\|_\infty = o_p(1). \quad (42)$$

Step 2. For $i, j = 1, \dots, K$, recall that $\varepsilon_i = (e_{i1}, \dots, e_{in})^\top$, and our goal is to prove that

$$\frac{1}{\sigma_i \|\hat{Z}_j\|_2} \sum_{t=1}^n \hat{Z}_{tj} e_{it} \xrightarrow{d} \mathcal{N}(0, 1). \quad (43)$$

Firstly, we prove the asymptotic normality of the sum of martingale difference sequence $n^{-1/2} \sum_{t=1}^n Z_{tj} e_{it}$. Let $\mathcal{A}_t := \sigma(\mathbf{e}_0, \dots, \mathbf{e}_t)$, then $Z_{tj} e_{it}$ is $\mathcal{A}_t$ -measurable, and $\mathcal{A}_{t-1}$ is contained in $\mathcal{A}_t$. As $Z_{tj} = X_{tj} - \sum_{l \neq j} \gamma_{jl} X_{tl}$ is $\mathcal{A}_{t-1}$ -measurable, we have

$$E(Z_{tj} e_{it} | \mathcal{A}_{t-1}) = 0. \quad (44)$$

Let $V_n^2 := \sum_{t=1}^n E(n^{-1} Z_{tj}^2 e_{it}^2 | \mathcal{A}_{t-1}) = \sigma_i^2 n^{-1} \sum_{t=1}^n Z_{tj}^2$ and $v_n^2 := E(V_n^2) = \sigma_i^2 \tau_j^2$. By (40), we have

$$V_n^2 v_n^{-2} \xrightarrow{P} 1. \quad (45)$$

As the elements of $\mathbf{X}$ are sub-Gaussian random variables, we have $\|\mathbf{X}\|_\infty = O_p(\sqrt{\log(K)})$. By Cauchy-Schwarz inequality, we have

$$\begin{aligned} n^{-1-\nu/2} \sum_{t=1}^n E|Z_{tj}|^{2+\nu} &\leq \frac{\|Z_j\|_2^2}{n} \left(\frac{\|Z_j\|_\infty}{\sqrt{n}} \right)^\nu \\ &\leq \frac{\|Z_j\|_2^2}{n} \left(\frac{(1 + \|\gamma_j\|_1) \|\mathbf{X}\|_\infty}{\sqrt{n}} \right)^\nu \\ &= O_p \left[(q_j \log(K)/n)^{\nu/2} \right]. \end{aligned} \quad (46)$$

Then, we have, in probability,

$$\begin{aligned} v_n^{-2} n^{-1} \sum_{t=1}^n E(Z_{tj}^2 e_{it}^2 1_{\{|n^{-1/2} Z_{tj} e_{it}| \geq \rho v_n\}}) &\leq v_n^{-2-\nu} n^{-1-\nu/2} \rho^{-\nu} \sum_{t=1}^n E|Z_{tj} e_{it}|^{2+\nu} \\ &\leq v_n^{-2-\nu} n^{-1-\nu/2} \rho^{-\nu} L \sum_{t=1}^n E|Z_{tj}|^{2+\nu} \\ &\rightarrow 0, \end{aligned} \quad (47)$$

where the first inequality holds by making use of $1_{\{|n^{-1/2} Z_{tj} e_{it}| \geq \rho v_n\}}$, the second inequality holds because e_{it} is sub-Gaussian random variable and L is a constant,

and the third is because of (46) and Assumption 4. Together with (44), (45) and (47), the martingale central limit theorem (Theorem 5.3.4 in Fuller (1996)) implies

$$\frac{1}{\sqrt{n}\sigma_i\tau_j} \sum_{t=1}^n Z_{tj}e_{it} \xrightarrow{d} \mathcal{N}(0, 1). \quad (48)$$

Secondly, we have

$$\begin{aligned} \frac{1}{\sqrt{n}} \sum_{t=1}^n (\hat{Z}_{tj} - Z_{tj})e_{it} &= \frac{1}{\sqrt{n}} \sum_{t=1}^n \sum_{l \neq j} (\gamma_{jl} - \hat{\gamma}_{jl}) X_{tl} e_{it} \\ &\leq \sqrt{n} \|\gamma_j - \hat{\gamma}_j\|_1 \|\mathbf{X}^T \varepsilon_i / n\|_\infty \\ &= O_p(q_j \log(K) / \sqrt{n}) \end{aligned} \quad (49)$$

where the first equality is due to the definition, the second inequality is due to Hölder inequality, and the third equality is due to (20) and (4). By (41), we have

$$\frac{\tau_j}{\|\hat{Z}_j\|_2} \xrightarrow{P} 1. \quad (50)$$

By Slutsky theorem, (48), (49), (50) and Assumption 4, we obtain (43).

Step 3. By Theorem 1, we have $\hat{s}_i = O_p(s_i) = o_p(n)$. Then $(n - \hat{s}_i)\hat{\sigma}_i^2/n$ is asymptotically equivalent to $\hat{\sigma}_i^2$. We have

$$\begin{aligned} |(n - \hat{s}_i)\hat{\sigma}_i^2/n - \sigma_i^2| &= |(\hat{\varepsilon}_i^T \hat{\varepsilon}_i - n\sigma_i^2)/n| \\ &= |(\hat{\varepsilon}_i - \varepsilon_i)^T (\hat{\varepsilon}_i - \varepsilon_i)/n + 2\varepsilon_i^T (\hat{\varepsilon}_i - \varepsilon_i)/n \\ &\quad + (\varepsilon_i^T \varepsilon_i - n\sigma_i^2)/n| \\ &= \|\mathbf{X}(\hat{a}_i^{\text{Lasso}} - a_i)\|_2^2/n - 2\varepsilon_i^T \mathbf{X}(\hat{a}_i^{\text{Lasso}} - a_i)/n \\ &\quad + (\varepsilon_i^T \varepsilon_i - n\sigma_i^2)/n| \\ &= O_p(s_i \log(K)/n) + O_p(\sqrt{\log(K)/n}) O_p(s_i \log(K)/n) \\ &\quad + O_p(1/\sqrt{n}) \\ &= o_p(1), \end{aligned}$$

where the first three equalities are due to definitions and simple algebra, the fourth equality is due to Theorem 1, Hölder inequality, (4) and the central limit theorem ($\varepsilon_i^T \varepsilon_i - n\sigma_i^2 = O_p(\sqrt{n})$), and the fifth equality is due to Assumption 3.

By Theorem 1 ($\hat{s}_i = O_p(s_i) = o_p(n)$), we obtain

$$\hat{\sigma}_i^2 - \sigma_i^2 = o_p(1). \quad (51)$$

55 Combining (42), (43) and (51), the results of Theorem 2 hold. $\square$

1.2. Proofs of the theoretical results in Section 3.2

We denote $\hat{\varepsilon}_{\text{cent}} := (\varepsilon_{\text{cent},1}, \dots, \varepsilon_{\text{cent},n})$, where $\varepsilon_{\text{cent},t} := \hat{\varepsilon}_{it} - \hat{\varepsilon}_{i\cdot}$. Note that we omit the subscript i for simplicity. When we are conditional on $\mathbf{X}$ and Y_i (therefore $\hat{\varepsilon}_{\text{cent}}$), the only randomness comes from ε_i^* which is generated from
60 residual bootstrap or multiplier wild bootstrap. As both bootstrap methods have a lot in common during the proof, we use the same symbol ε_i^* to refer to the bootstrap residuals and discuss them separately when necessary.

Proposition 4. (a) Under Assumptions 1 and 6, we have

$$\|\mathbf{X}\|_{\infty} = O_p(\sqrt{\log(K)}). \quad (52)$$

(b) Under Assumptions 1 and 6, for tuning parameter satisfying $\lambda \asymp \sqrt{\log(K)/n}$, we have

$$E^*(\|\varepsilon_i^*\|_{2+\phi}^{2+\phi}/n) = O_p(1), \quad (53)$$

where $0 \leq \phi < 1$. In particular, when $\phi = 0$, we have

$$E^*(\|\varepsilon_i^*\|_2^2/n) = O_p(1). \quad (54)$$

Proof. The result of (a) follows from that the elements of $\mathbf{X}$ are sub-Gaussian random variables.

As $\varepsilon_i = (e_{i1}, \dots, e_{in})^T$ is sub-Gaussian distributed, we have $E\|\varepsilon_i\|_{2+\phi}^{2+\phi} = O(n)$. We denote $\varepsilon_{i\cdot} := \sum_{t=1}^n e_{it}/n$, then

$$|\hat{\varepsilon}_{i\cdot}| \leq |\hat{\varepsilon}_{i\cdot} - \varepsilon_{i\cdot}| + |\varepsilon_{i\cdot}| \leq \|\hat{\varepsilon}_i - \varepsilon_i\|_1/n + |\varepsilon_{i\cdot}| = o_p(1)$$

where the first two inequality are due to triangle inequality and the third is due to Theorem 1, Assumption 6 and the strong law of larger numbers. By Theorem 1, (52) and Assumption 6, we have

$$\|\hat{\varepsilon}_{\text{cent}} - \varepsilon_i\|_{\infty} \leq |\hat{\varepsilon}_{i\cdot}| + \|\hat{\varepsilon}_i - \varepsilon_i\|_{\infty} \leq |\hat{\varepsilon}_{i\cdot}| + \|\mathbf{X}\|_{\infty} \|\hat{a}_i^{\text{Lasso}} - a_i\|_1 = o_p(1).$$

Thus, we have

$$\|\hat{\varepsilon}_{\text{cent}}\|_{2+\phi}^{2+\phi}/n = O_p(1).$$

For the residual bootstrap,

$$E^*(\|\varepsilon_i^*\|_{2+\phi}^{2+\phi}/n) = \|\hat{\varepsilon}_{\text{cent}}\|_{2+\phi}^{2+\phi}/n = O_p(1).$$

For the multiplier wild bootstrap,

$$E^*(\|\varepsilon_i^*\|_{2+\phi}^{2+\phi}/n) = (E|W_{11}|^{2+\phi}) \|\hat{\varepsilon}_{\text{cent}}\|_{2+\phi}^{2+\phi}/n = O_p(1).$$

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□

Proof of Theorem 3 (a). Recall that we intend to prove that

$$\begin{aligned} \|\hat{a}_i^{\text{Lasso}^*} - \hat{a}_i^{\text{Lasso}}\|_1 &= O_p(s_i \log(K)/\sqrt{n}), \\ \|\mathbf{X}(\hat{a}_i^{\text{Lasso}^*} - \hat{a}_i^{\text{Lasso}})\|_2^2/n &= O_p(s_i \log^2(K)/n). \end{aligned}$$

By Theorem 1, we have $\hat{s}_i = O_p(s_i)$, thus, we only need to prove that

$$\|\hat{a}_i^{\text{Lasso}^*} - \hat{a}_i^{\text{Lasso}}\|_1 = O_p(\hat{s}_i \log(K)/\sqrt{n}), \quad (55)$$

$$\|\mathbf{X}(\hat{a}_i^{\text{Lasso}^*} - \hat{a}_i^{\text{Lasso}})\|_2^2/n = O_p(\hat{s}_i \log^2(K)/n). \quad (56)$$

Recall that

$$\hat{a}_i^{\text{Lasso}^*} := \underset{\alpha \in \mathbb{R}^K}{\operatorname{argmin}} \{ \|Y_i^* - \mathbf{X}\alpha\|_2^2/n + 2\lambda^* \|\alpha\|_1 \},$$

then we obtain the basic inequality

$$\|Y_i^* - \mathbf{X}\hat{a}_i^{\text{Lasso}^*}\|_2^2/n + 2\lambda^* \|\hat{a}_i^{\text{Lasso}^*}\|_1 \leq \|Y_i^* - \mathbf{X}\hat{a}_i^{\text{Lasso}}\|_2^2/n + 2\lambda^* \|\hat{a}_i^{\text{Lasso}}\|_1.$$

We denote $\delta^* := \hat{a}_i^{\text{Lasso}^*} - \hat{a}_i^{\text{Lasso}}$. By simple algebra, we have

$$\|\mathbf{X}\delta^*\|_2^2/n \leq 2\varepsilon_i^{*\text{T}}\mathbf{X}\delta^*/n + 2\lambda^*(\|\hat{a}_i^{\text{Lasso}}\|_1 - \|\hat{a}_i^{\text{Lasso}^*}\|_1). \quad (57)$$

We can obtain a bound for $\|\mathbf{X}^T \varepsilon_i^*/n\|_\infty$ as follows:

$$P^* \left(\|\mathbf{X}^T \varepsilon_i^*/n\|_\infty > C_2 \frac{\log(K)}{\sqrt{n}} \right) \leq \frac{E^*(\|\mathbf{X}^T \varepsilon_i^*\|_\infty)^2}{C_2 n \log^2(K)} \leq \frac{8 \log(2K) \|\mathbf{X}\|_\infty^2 E^*(\|\varepsilon_i^*\|_2^2)}{C_2 n \log^2 K},$$

where the first inequality is due to Markov inequality, and the second inequality is due to Nemirovski's inequality and equation (6.5) in Bühlmann and van de Geer (2011). By (52) and (54), we have, in probability,

$$\|\mathbf{X}^T \varepsilon_i^*/n\|_\infty \leq C_2 \frac{\log(K)}{\sqrt{n}}. \quad (58)$$

With suitable chosen $\lambda^* \asymp \log(K)/\sqrt{n}$, we have, in probability,

$$\|\mathbf{X}^T \varepsilon_i^*/n\|_\infty \leq C_2 \frac{\log(K)}{\sqrt{n}} \leq \lambda^*/2.$$

Furthermore, by Hölder inequality, we have

$$\varepsilon_i^{*T} \mathbf{X} \delta^*/n \leq \|\mathbf{X}^T \varepsilon_i^*/n\|_\infty \|\delta^*\|_1 \leq \lambda^* \|\delta^*\|_1/2.$$

Recall that $\hat{S}_i = \{j \in \{1, \dots, K\} : \hat{a}_{ij}^{\text{Lasso}} \neq 0\}$, by triangle inequality and $\hat{a}_{i\hat{S}_i^C}^{\text{Lasso}} = 0$, we have

$$\begin{aligned} \|\hat{a}_i^{\text{Lasso}}\|_1 - \|\hat{a}_i^{\text{Lasso}*}\|_1 &= \|\hat{a}_{i\hat{S}_i}^{\text{Lasso}}\|_1 - \|\hat{a}_{i\hat{S}_i}^{\text{Lasso}*}\|_1 + \|\hat{a}_{i\hat{S}_i^C}^{\text{Lasso}}\|_1 - \|\hat{a}_{i\hat{S}_i^C}^{\text{Lasso}*}\|_1 \\ &\leq \|\delta_{\hat{S}_i}^*\|_1 - \|\delta_{\hat{S}_i^C}^*\|_1. \end{aligned}$$

Therefore, (57) becomes

$$\begin{aligned} 0 &\leq \|\mathbf{X} \delta^*\|_2^2/n \leq \lambda^* \|\delta^*\|_1 + 2\lambda^* (\|\delta_{\hat{S}_i}^*\|_1 - \|\delta_{\hat{S}_i^C}^*\|_1) \\ &= \lambda^* (\|\delta_{\hat{S}_i}^*\|_1 + \|\delta_{\hat{S}_i^C}^*\|_1) + 2\lambda^* (\|\delta_{\hat{S}_i}^*\|_1 - \|\delta_{\hat{S}_i^C}^*\|_1) \\ &= 3\lambda^* \|\delta_{\hat{S}_i}^*\|_1 - \lambda^* \|\delta_{\hat{S}_i^C}^*\|_1 \\ &\leq 3\lambda^* \|\delta^*\|_1. \end{aligned} \quad (59)$$

Inequality (59) implies $\|\delta_{\hat{S}_i^C}^*\|_1 \leq 3\|\delta_{\hat{S}_i}^*\|_1$ so that

$$\|\delta^*\|_1 \leq 4\|\delta_{\hat{S}_i}^*\|_1 \leq 4\sqrt{\hat{s}_i} \|\delta^*\|_2. \quad (60)$$

By (3) and (24), we have

$$\|\mathbf{X} \delta^*\|_2^2/n \geq \alpha \|\delta\|_2^2 - \tau \|\delta\|_1^2 \geq \left(\frac{\alpha}{4\hat{s}_i} - \tau \right) \|\delta\|_1^2. \quad (61)$$

According to Theorem 1 and Proposition 3, we have $\tau \hat{s}_i = O_p(\tau s_i) = O_p(1)$. Combined with (59), (61) and Assumption 5 ($\lambda^* \asymp \log(K)/\sqrt{n}$), we obtain (55). Then by (59), (55) and Assumption 5, we obtain (56). $\square$

Proof of Theorem 3 (b). By Theorem 1, we have $\hat{s}_i = O_p(s_i)$. To prove
70 $\hat{s}_i^* = O_p(s_i)$, we only need to prove $\hat{s}_i^* = O_p(\hat{s}_i)$.

The proof is the same as the counterpart in Theorem 1 (b) except for the bound (17). We only need to replace

$$\|\mathbf{X}^T \varepsilon_i / n\|_\infty \leq C_1 \sqrt{\log(K)/n},$$

by

$$\|\mathbf{X}^T \varepsilon_i^* / n\|_\infty \leq C_2 \log(K) / \sqrt{n},$$

where the second inequality is due to (58). With

$$\lambda^* \geq \frac{4C_2 c^* \log(K)}{c_* \sqrt{n}},$$

the result follows from the same reasoning. $\square$

Similar to (26), we have the bootstrap analogue,

$$\hat{a}_i^* - \hat{a}_i^{\text{Lasso}} = \hat{\Theta} \mathbf{X}^T \varepsilon_i^* / n + (I - \hat{\Theta} \hat{\Sigma})(\hat{a}_i^{\text{Lasso}*} - \hat{a}_i^{\text{Lasso}}).$$

As we use the same $\mathbf{X}$ for the nodewise Lasso during the bootstrap procedure, the bounds proved in the proof of Theorem 2 still hold.

Proof of Theorem 4. We prove the theorem in three steps. Step 1 proves
75 that $(I - \hat{\Theta} \hat{\Sigma})(\hat{a}_i^{\text{Lasso}*} - \hat{a}_i^{\text{Lasso}})$ is asymptotically negligible. Step 2 proves that $\hat{\Theta} \mathbf{X}^T \varepsilon_i^* / n$ is asymptotically normal. Step 3 proves that our variance estimator is consistent.

Step 1. By Hölder inequality, we have

$$\begin{aligned} \sqrt{n} \|(I - \hat{\Theta} \hat{\Sigma})(\hat{a}_i^{\text{Lasso}*} - \hat{a}_i^{\text{Lasso}})\|_\infty &\leq \sqrt{n} \|(I - \hat{\Theta} \hat{\Sigma})\|_\infty \|\hat{a}_i^{\text{Lasso}*} - \hat{a}_i^{\text{Lasso}}\|_1 \\ &\leq \sqrt{n} \left(\max_j \lambda_j / \hat{\tau}_j^2 \right) \|\hat{a}_i^{\text{Lasso}*} - \hat{a}_i^{\text{Lasso}}\|_1. \end{aligned} \tag{62}$$

According to Theorem 3, we have

$$\|\hat{a}_i^{\text{Lasso}*} - \hat{a}_i^{\text{Lasso}}\|_1 = O_p(s_i \log(K) / \sqrt{n}). \tag{63}$$

As we use the same $\mathbf{X}$ for the nodewise Lasso during the bootstrap procedure, (32) still holds. Together with Assumptions 5 and 6, we have

$$\sqrt{n}\|(I - \hat{\Theta}\hat{\Sigma})(\hat{a}_i^{\text{Lasso}*} - \hat{a}_i^{\text{Lasso}})\|_\infty = o_p(1). \quad (64)$$

Step 2. Compared to the counterpart in Theorem 2, $\hat{Z}_{ij}$ is no longer correlated with ε_i^* when we conditional on $\mathbf{X}$ and consider the bootstrap measure. By (53),

$$\left\{ \frac{1}{\|\hat{\varepsilon}_{\text{cent}}\|_2^2/n \cdot \|\hat{Z}_j\|_2} \hat{Z}_{tj} \varepsilon_{1t}^*, t = 1, \dots, n \right\}$$

are independent random variables meeting the Lyapunov condition. By central limit theorem,

$$\frac{1}{\|\hat{\varepsilon}_{\text{cent}}\|_2^2/n \cdot \|\hat{Z}_j\|_2} \sum_{t=1}^n \hat{Z}_{tj} \varepsilon_{1t}^* \xrightarrow{d^*} \mathcal{N}(0, 1) \text{ in probability.} \quad (65)$$

Step 3. By Theorem 3, we have $\hat{s}_i^* = O_p(s_i) = o_p(n)$. Then $(n - \hat{s}_i^*)\hat{\sigma}_i^{*2}/n$ is asymptotically equivalent to $\hat{\sigma}_i^{*2}$. Moreover, we have

$$\begin{aligned} & |(n - \hat{s}_i^*)\hat{\sigma}_i^{*2}/n - \|\hat{\varepsilon}_{\text{cent}}\|_2^2/n| \\ &= |(\hat{\varepsilon}_i^{*\text{T}} \hat{\varepsilon}_i^* - \|\hat{\varepsilon}_{\text{cent}}\|_2^2)/n| \\ &= |(\hat{\varepsilon}_i^* - \varepsilon_i^*)^{\text{T}}(\hat{\varepsilon}_i^* - \varepsilon_i^*)/n + 2\varepsilon_i^{*\text{T}}(\hat{\varepsilon}_i^* - \varepsilon_i^*)/n + (\varepsilon_i^{\text{T}} \varepsilon_i^* - \|\hat{\varepsilon}_{\text{cent}}\|_2^2)/n| \\ &= \|\mathbf{X}(\hat{a}_i^{\text{Lasso}*} - \hat{a}_i^{\text{Lasso}})\|_2^2/n - 2\varepsilon_i^{*\text{T}}\mathbf{X}(\hat{a}_i^{\text{Lasso}*} - \hat{a}_i^{\text{Lasso}})/n + (\varepsilon_i^{*\text{T}} \varepsilon_i^* - \|\hat{\varepsilon}_{\text{cent}}\|_2^2)/n \\ &= O_p(s_i \log^2(K)/n) + O_p(\log(K)/\sqrt{n}) O_p(s_i \log(K)/\sqrt{n}) + O_p(1/\sqrt{n}) \\ &= o_p(1), \end{aligned}$$

where the first three equalities are due to definitions and simple algebra, the fourth equality is due to Theorem 3, (58), central limit theorem, and (54), and the fifth equality is due to Assumption 6. According to Theorem 3 ($\hat{s}_i^* = O_p(s_i) = o_p(n)$), we obtain,

$$\hat{\sigma}_i^{*2} - \|\hat{\varepsilon}_{\text{cent}}\|_2^2/n = o_p(1) \text{ in probability.} \quad (66)$$

Combining (64), (65) and (66), the results of Theorem 4 follow. $\square$

2. Additional numerical results

80 2.1. Additional cases

Tables 1 and 2 shows the additional results for t distributed noise when $n = 150$. The overall performance is similar to the case of $n = 100$.

Tables 3 - 6 show the performance of proposed methods in the case with Gaussian and sub-Gaussian noise. Firstly, we compare the absolute bias and
85 RMSE (Figures 1–2). For non-zero parameters, the Lasso estimator has large finite-sample biases, while the Lasso+OLS estimator reduces the bias by 41% - 68%, JM reduces the bias by 74% - 84%, and LDPE reduces the bias by 82% - 85%. For zero parameters, Lasso and Lasso+OLS have near-zero biases while JM and LDPE have small but non-zero finite-sample biases. In terms of RMSE,
90 compared to LDPE and JM, Lasso and Lasso+OLS have smaller RMSE for those zero parameters but larger RMSE for those non-zero parameters. As the number of zero parameters is much larger than that of non-zero parameters, Lasso and Lasso+OLS have smaller overall RMSE, 73% - 81% smaller than those of LDPE and JM. Therefore, for estimation purposes, we recommend the
95 Lasso and Lasso+OLS because of their smaller overall RMSEs. However, for the construction of confidence intervals, the two de-biased Lasso methods work better than the Lasso and Lasso+OLS because they have smaller biases for all parameters, which usually lead to more accurate coverage probability.

Secondly, we compare the coverage probabilities and average confidence interval lengths of 95% confidence intervals (Figures 3–4). For zero parameters, all
100 methods reach the nominal confidence level. In contrast, for non-zero parameters, the coverage probabilities of BtLasso, BtLasso+OLS, and JM fail to reach the nominal confidence level, while the LDPE, BtLDPE, MultiBtLDPE have good coverage probabilities in most cases. Moreover, compared to the LDPE,
105 the BtLDPE and the MultiBtLDPE reduce the interval length by about 3%. Our simulation results confirm our theory on the validity of LDPE, BtLDPE, and MultiBtLDPE for statistical inference.

Table 1: Performance of different methods in the case of t_1 distributed noise and $n = 150$.

type	method	bias	rmse	cp	length	time
non-zero	BtLasso	2(0)	3(0)	55(0)	9(0)	-
	BtLassoOLS	1(0)	3(1)	64(0)	16(1)	-
	JM	4(3)	96(51)	68(1)	44(7)	-
	LDPE	2(1)	39(18)	84(0)	77(13)	-
	BtLDPE	2(1)	39(18)	87(0)	69(10)	-
	MultiBtLDPE	2(1)	39(17)	95(0)	78(5)	-
	ModelBtLDPE	2(1)	22(6)	23(0)	104(25)	-
zero	BtLasso	0(0)	10(2)	96(0)	5(1)	-
	BtLassoOLS	1(0)	25(4)	96(0)	13(2)	-
	JM	94(69)	2095(1312)	94(0)	72(13)	-
	LDPE	70(63)	1554(1059)	91(0)	122(25)	-
	BtLDPE	70(67)	1554(1051)	92(0)	114(27)	-
	MultiBtLDPE	70(61)	1566(1017)	97(0)	67(12)	-
	ModelBtLDPE	82(75)	816(563)	75(1)	182(76)	-
overall	BtLasso	2(0)	11(2)	95(0)	5(1)	86
	BtLassoOLS	2(0)	25(3)	95(0)	13(2)	173
	JM	94(80)	2098(1319)	93(0)	71(14)	21
	LDPE	70(62)	1555(1027)	91(0)	120(23)	27
	BtLDPE	70(64)	1555(1060)	92(0)	113(26)	196
	MultiBtLDPE	70(62)	1566(1030)	97(0)	67(12)	276
	ModelBtLDPE	82(72)	816(548)	73(1)	180(71)	2808

bias indicates the absolute bias; rmse is the root-mean-squared error; cp is the empirical coverage probability; length is the mean interval length; time indicates the CPU hours required for 500 simulation. The numbers in brackets are the corresponding standard errors estimated using the bootstrap with 200 replications. cp, length, and their standard errors are multiplied by 100.

Table 2: Performance of different methods in the case of t_2 distributed noise and $n = 150$.

type	method	bias	rmse	cp	length	time
non-zero	BtLasso	2(0)	3(0)	51(0)	13(0)	-
	BtLassoOLS	1(0)	2(0)	74(0)	22(0)	-
	JM	0(0)	2(0)	93(0)	33(0)	-
	LDPE	0(0)	2(0)	94(0)	33(0)	-
	BtLDPE	0(0)	2(0)	93(0)	32(0)	-
	MultiBtLDPE	0(0)	2(0)	97(0)	44(0)	-
	ModelBtLDPE	1(0)	2(0)	27(0)	33(0)	-
zero	BtLasso	0(0)	2(0)	95(0)	3(0)	-
	BtLassoOLS	0(0)	4(0)	95(0)	9(0)	-
	JM	1(0)	12(1)	95(0)	34(0)	-
	LDPE	1(0)	12(1)	94(0)	35(0)	-
	BtLDPE	1(0)	12(1)	93(0)	33(0)	-
	MultiBtLDPE	1(0)	12(1)	95(0)	30(0)	-
	ModelBtLDPE	2(0)	12(1)	68(0)	34(0)	-
overall	BtLasso	2(0)	3(0)	94(0)	3(0)	26
	BtLassoOLS	1(0)	5(0)	95(0)	10(0)	55
	JM	1(0)	12(1)	95(0)	34(0)	15
	LDPE	1(0)	12(1)	94(0)	35(0)	11
	BtLDPE	1(0)	12(1)	93(0)	33(0)	80
	MultiBtLDPE	1(0)	12(1)	95(0)	30(0)	63
	ModelBtLDPE	2(0)	12(1)	67(0)	34(0)	2785

bias indicates the absolute bias; rmse is the root-mean-squared error; cp is the empirical coverage probability; length is the mean interval length; time indicates the CPU hours required for 500 simulation. The numbers in brackets are the corresponding standard errors estimated using the bootstrap with 200 replications. cp, length, and their standard errors are multiplied by 100.

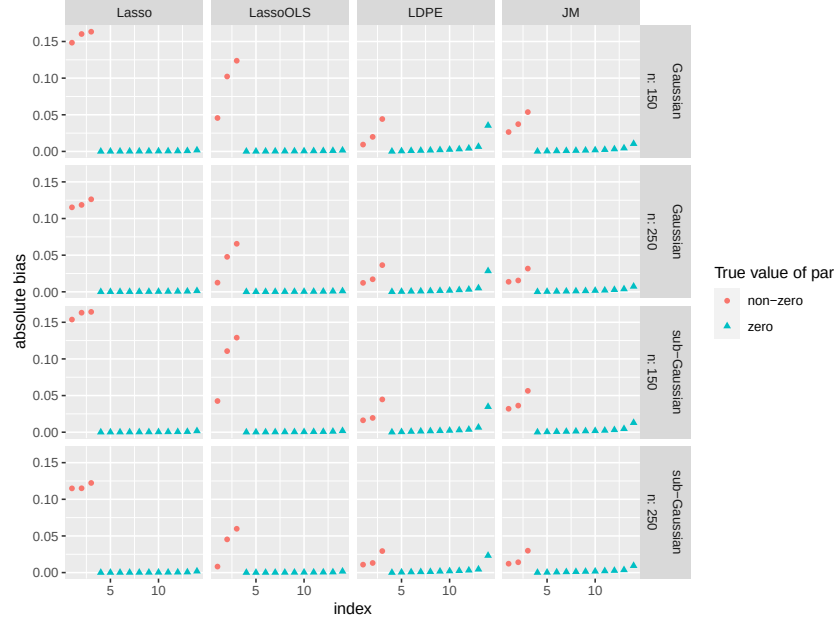


Figure 1: Comparison of absolute biases of four methods (columns of panels) in four cases (rows of panels). Index on the x -axis corresponds to different a_{1j} 's. Three non-zero a_{1j} 's are sorted by absolute bias and displayed. The remaining 197 zero a_{1j} are also sorted by absolute bias, and then ten of them are selected equidistantly and displayed.

2.2. Alternative methods for tuning parameter selection and variance estimation

This section examines the statistical performance of alternative methods for
110 tuning parameter selection and variance estimation. We compare the methods
in the case of t_2 noise and $n = 100$.

In addition to regular cross-validation (LDPEcv), two alternative methods
are used for tuning parameter selection. One is the time-series cross-validation
(LDPEts cv), which divides the time series data into ten consecutive folds. The
115 other is based on Bayes Information Criterion (LDPEbic), using Bayes Informa-
tion Criterion to select the best tuning parameter. Table 7 shows that LDPEcv
outperforms the other two methods.

We also examine the performance of the variance estimator without cor-
rection of the degree of freedom (LDPEsig). Table 7 shows that the coverage

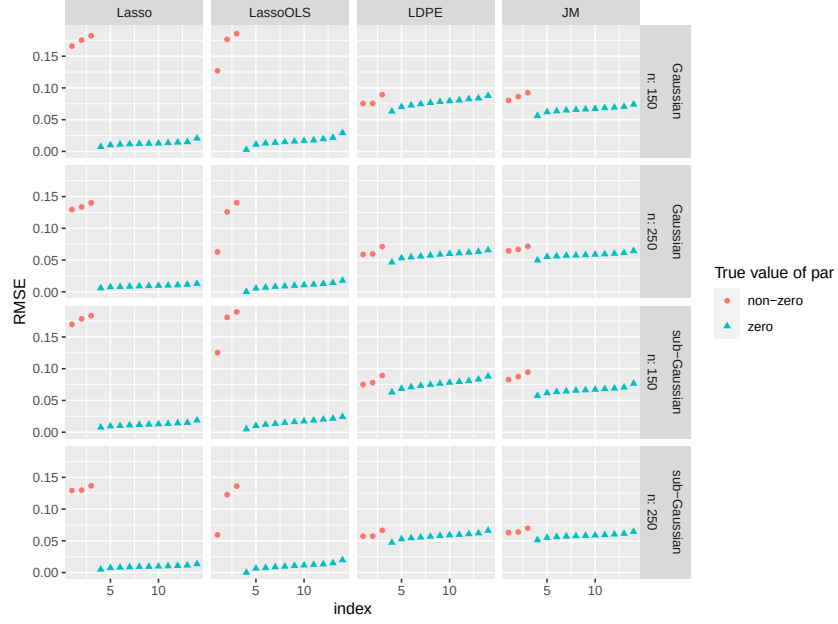


Figure 2: Comparison of RMSE for 1000 replications produced by four methods (columns of panels) in four cases (rows of panels). Index on the x -axis corresponds to different a_{1j} 's. All three non-zero a_{1j} are sorted by RMSE and displayed. All 197 zero a_{1j} are also sorted by RMSE. Then ten zero a_{1j} are selected equidistantly and displayed.

120 probabilities of the LDPEsig are lower than that of LDPEcv. Thus, this variance estimator does not work well.

2.3. Details of real data analysis

In this section, we shows the results of BtLDPE + WY on the S&P 500 data set. Figure 5 shows all significant edges and Table 8 shows the detailed
125 information of the top 10 significant edges among them.

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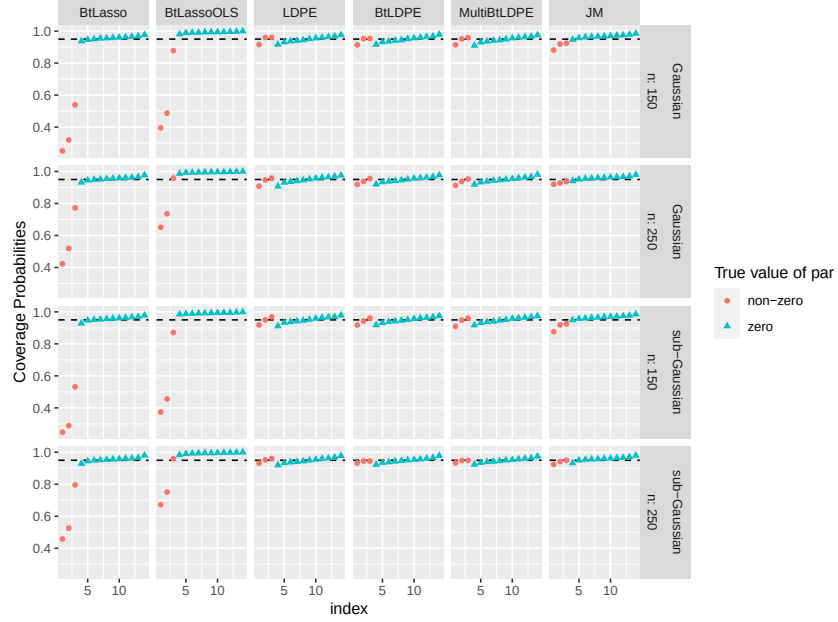


Figure 3: Comparison of empirical coverage probabilities of 95% confidence intervals produced by six methods (columns of panels) in four cases (rows of panels). Index on the x -axis corresponds to different a_{1j} 's. Three non-zero a_{1j} are sorted by empirical coverage probabilities and displayed. The remaining 197 zero a_{1j} are also sorted by empirical coverage probabilities, and then ten of them are selected equidistantly and displayed. The dashed lines correspond to the nominal confidence level 95%.

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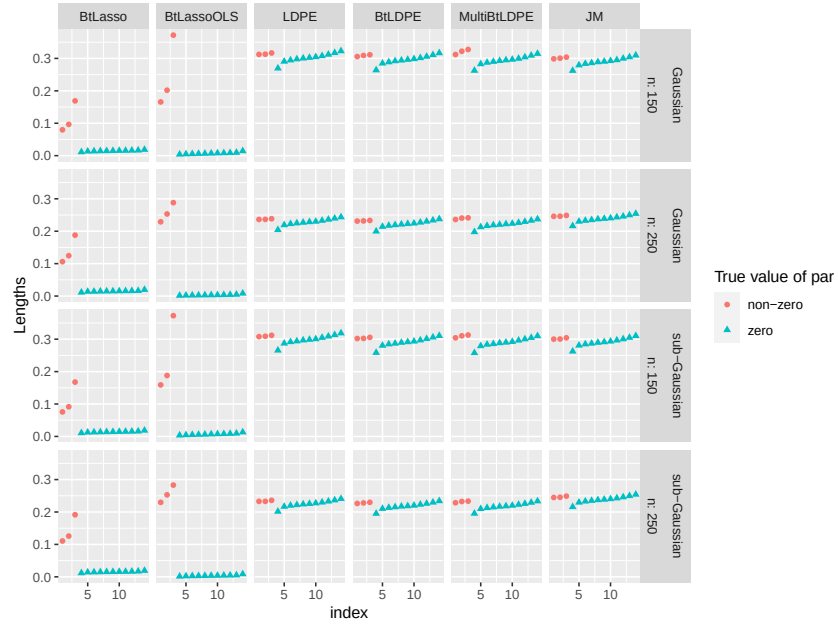


Figure 4: Comparison of average confidence interval lengths of 95% confidence intervals produced by six methods (columns of panels) in four cases (rows of panels). Index on the x -axis corresponds to different a_{1j} 's. Three non-zero a_{1j} are sorted by interval lengths and displayed. The remaining 197 zero a_{1j} are also sorted by interval lengths, and then ten of them are selected equidistantly and displayed.

Table 3: Simulated absolute biases, RMSEs, coverage probabilities, and average interval lengths of different methods in the case of Gaussian noise and $n = 150$. The nominal confidence level is 95%. The numbers in parentheses are simulated standard errors. All numbers are multiplied by 1000.

type	method	bias	RMSE	cp	length
non-zero	BtLasso	157(1)	302(3)	370(9)	115(1)
	BtLassoOLS	91(2)	286(3)	587(8)	246(3)
	JM	39(1)	150(2)	908(5)	301(1)
	LDPE	24(1)	139(2)	946(4)	314(0)
	BtLDPE	25(1)	139(2)	940(4)	309(1)
	MultiBtLDPE	24(1)	139(2)	941(4)	321(1)
zero	BtLasso	0(0)	176(5)	957(0)	15(0)
	BtLassoOLS	0(0)	231(6)	993(0)	7(0)
	JM	2(0)	930(2)	968(0)	291(0)
	LDPE	3(0)	1087(3)	950(0)	302(0)
	BtLDPE	3(0)	1087(3)	949(0)	296(0)
	MultiBtLDPE	3(0)	1088(3)	949(0)	295(0)
overall	BtLasso	3(0)	350(3)	949(0)	16(0)
	BtLassoOLS	2(0)	367(4)	987(0)	11(0)
	JM	2(0)	942(2)	967(0)	291(0)
	LDPE	4(0)	1096(3)	950(0)	303(0)
	BtLDPE	4(0)	1096(3)	949(0)	296(0)
	MultiBtLDPE	4(0)	1096(3)	949(0)	295(0)

Table 4: Simulated absolute biases, RMSEs, coverage probabilities, and average interval lengths of different methods in the case of sub-Gaussian noise and $n = 150$. The nominal confidence level is 95%. The numbers in parentheses are simulated standard errors. All numbers are multiplied by 1000.

type	method	bias	RMSE	cp	length
non-zero	BtLasso	160(1)	307(3)	356(8)	112(1)
	BtLassoOLS	94(2)	290(3)	567(8)	240(3)
	JM	42(1)	153(2)	907(5)	302(1)
	LDPE	27(1)	140(2)	946(4)	310(0)
	BtLDPE	27(1)	140(2)	941(4)	303(0)
	MultiBtLDPE	27(1)	140(2)	939(4)	309(1)
zero	BtLasso	0(0)	174(4)	958(0)	14(0)
	BtLassoOLS	0(0)	230(6)	992(0)	7(0)
	JM	2(0)	930(1)	967(0)	292(0)
	LDPE	3(0)	1069(2)	951(0)	299(0)
	BtLDPE	3(0)	1070(2)	950(0)	291(0)
	MultiBtLDPE	3(0)	1069(2)	950(0)	290(0)
overall	BtLasso	3(0)	353(3)	949(0)	16(0)
	BtLassoOLS	2(0)	370(4)	986(0)	10(0)
	JM	3(0)	943(1)	966(0)	292(0)
	LDPE	3(0)	1078(2)	951(0)	299(0)
	BtLDPE	3(0)	1079(2)	950(0)	292(0)
	MultiBtLDPE	3(0)	1078(2)	950(0)	291(0)

Table 5: Simulated absolute biases, RMSEs, coverage probabilities, and average interval lengths of different methods in the case of Gaussian noise and $n = 250$. The nominal confidence level is 95%. The numbers in parentheses are simulated standard errors. All numbers are multiplied by 1000.

type	method	bias	RMSE	cp	length
non-zero	BtLasso	120(1)	233(2)	571(9)	139(1)
	BtLassoOLS	42(2)	199(3)	782(7)	257(3)
	JM	20(1)	117(1)	928(5)	247(0)
	LDPE	22(1)	110(1)	938(4)	237(0)
	BtLDPE	22(1)	110(1)	938(4)	232(0)
	MultiBtLDPE	22(1)	110(1)	935(5)	239(0)
zero	BtLasso	0(0)	132(3)	956(0)	15(0)
	BtLassoOLS	0(0)	142(4)	996(0)	3(0)
	JM	2(0)	816(2)	961(0)	239(0)
	LDPE	2(0)	819(2)	950(0)	228(0)
	BtLDPE	2(0)	819(2)	950(0)	223(0)
	MultiBtLDPE	2(0)	819(2)	950(0)	222(0)
overall	BtLasso	2(0)	268(2)	950(0)	17(0)
	BtLassoOLS	1(0)	244(3)	993(0)	7(0)
	JM	2(0)	825(2)	960(0)	239(0)
	LDPE	3(0)	826(2)	950(0)	229(0)
	BtLDPE	3(0)	826(2)	950(0)	223(0)
	MultiBtLDPE	3(0)	826(2)	950(0)	222(0)

Table 6: Simulated absolute biases, RMSEs, coverage probabilities, and average interval lengths of different methods in the case of sub-Gaussian noise and $n = 250$. The nominal confidence level is 95%. The numbers in parentheses are simulated standard errors. All numbers are multiplied by 1000.

type	method	bias	RMSE	cp	length
non-zero	BtLasso	117(1)	229(2)	593(9)	143(1)
	BtLassoOLS	38(2)	193(3)	794(7)	255(2)
	JM	19(1)	114(1)	939(4)	246(0)
	LDPE	18(1)	105(1)	948(4)	234(0)
	BtLDPE	18(1)	105(1)	941(4)	228(0)
	MultiBtLDPE	18(1)	105(1)	944(4)	232(0)
zero	BtLasso	0(0)	134(2)	955(0)	15(0)
	BtLassoOLS	0(0)	154(4)	995(0)	4(0)
	JM	2(0)	816(1)	960(0)	239(0)
	LDPE	2(0)	808(1)	950(0)	226(0)
	BtLDPE	2(0)	808(1)	950(0)	219(0)
	MultiBtLDPE	2(0)	808(1)	950(0)	218(0)
overall	BtLasso	2(0)	265(2)	950(0)	17(0)
	BtLassoOLS	1(0)	247(3)	992(0)	7(0)
	JM	2(0)	824(1)	960(0)	239(0)
	LDPE	3(0)	815(1)	950(0)	226(0)
	BtLDPE	3(0)	815(1)	950(0)	219(0)
	MultiBtLDPE	3(0)	815(1)	950(0)	219(0)

Table 7: Simulated absolute biases, RMSEs, coverage probabilities, and average interval lengths of alternative procedures in the case of t_2 distributed noise and $n = 100$. The nominal confidence level is 95%. The numbers in parentheses are simulated standard errors. All numbers are multiplied by 1000.

type	method	bias	RMSE	cp	length
non-zero	LDPEcv	32(2)	219(12)	935(4)	421(7)
	LDPEtscv	33(2)	223(12)	924(5)	420(7)
	LDPEbic	49(2)	214(7)	749(8)	278(11)
	LDPEsig	33(2)	223(12)	925(5)	420(7)
zero	LDPEcv	6(0)	2220(362)	946(1)	420(1)
	LDPEtscv	6(0)	2248(357)	936(1)	416(1)
	LDPEbic	5(0)	1951(313)	845(1)	276(2)
	LDPEsig	6(0)	2251(357)	936(1)	420(1)
overall	LDPEcv	6(0)	2231(360)	946(1)	420(1)
	LDPEtscv	6(0)	2259(356)	936(1)	416(1)
	LDPEbic	6(0)	1963(312)	844(1)	276(2)
	LDPEsig	6(0)	2262(355)	936(1)	420(1)

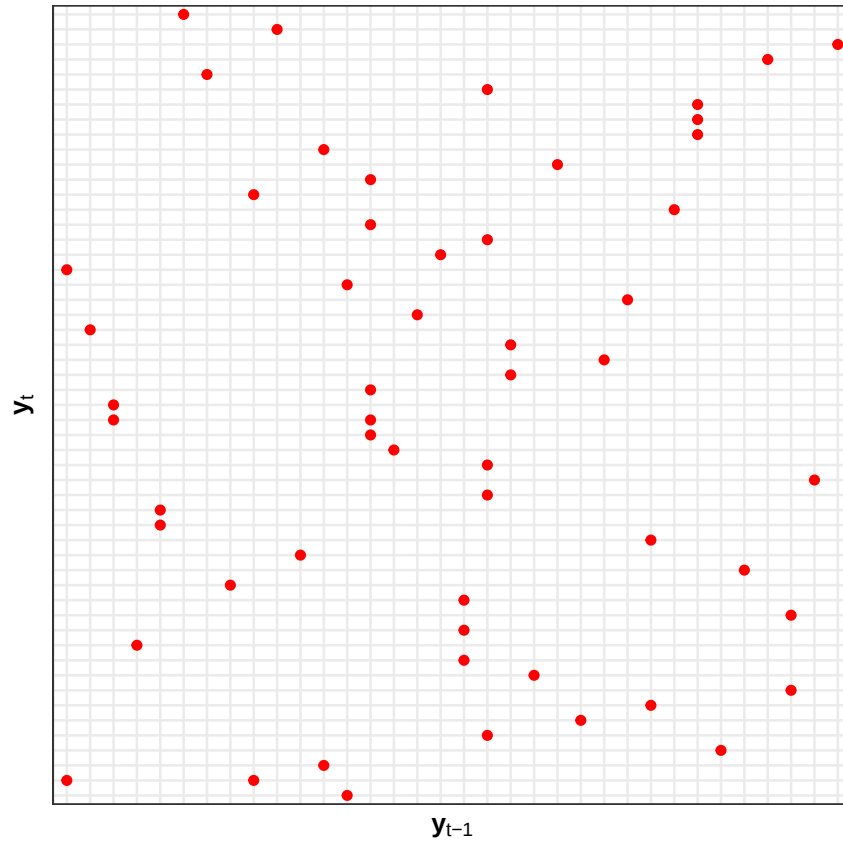


Figure 5: Significant edges of A using BtLDPE with Westfall and Young's multiple testing correction.

Table 8: Top 10 significant edges detected by BtLDPE with Westfall-Young procedure for multiple testing adjustment.

$\mathbf{y}_{t-1}$	$\mathbf{y}_t$	p -value
W. R. Berkley Corporation	Alliance Data Systems Corporation	0.004
Stryker Corporation	Charter Communications Inc. Class A	0.004
Stryker Corporation	Mondelez International Inc. Class A	0.004
Kraft Heinz Company	Danaher Corporation	0.004
Newmont Corporation	Eastman Chemical Company	0.004
Stryker Corporation	ServiceNow Inc.	0.006
Diamondback Energy Inc.	WEC Energy Group Inc	0.006
Diamondback Energy Inc.	Pinnacle West Capital Corporation	0.006
Texas Instruments Incorporated	Varian Medical Systems Inc.	0.008
Tiffany & Co.	Leggett & Platt Incorporated	0.008